
Learning Latent Sentiment Boundaries for Fine-Grained Multimodal Sentiment Analysis

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Abstract

1 Multimodal Sentiment Analysis (MSA) aims to infer human sentiment from lan-
2 guage, audio, and visual signals. Recent multimodal large language models
3 (MLLMs) have shown strong general-purpose multimodal understanding, creating
4 new opportunities for MSA. However, our preliminary study reveals that current
5 MLLMs still struggle with fine-grained affective recognition: they can often capture
6 coarse sentiment direction, but fail to reliably distinguish adjacent affective states
7 in an ordered label space. We further find that such errors are closely related to the
8 failure of forming stable gold-over-adjacent boundary advantages in middle-to-late
9 latent layers. To address this issue, we propose LSB-Distill, a Latent Sentiment
10 Boundary Distillation framework. LSB-Distill uses a reference-conditioned teacher
11 mode during training to provide layer-wise latent boundary targets, and distills them
12 into the standard student mode without adding inference-time cost. Experiments
13 on CH-SIMS, CMU-MOSI, and CMU-MOSEI show that LSB-Distill improves
14 fine-grained affective recognition and reduces adjacent-boundary errors across
15 representative MLLM backbones.

16 1 Introduction

17 Multimodal Sentiment Analysis (MSA) infers sentiment from language, audio, and visual cues. It
18 supports many applications, including human-computer interaction [15, 34], dialogue systems [24,
19 31], customer feedback analysis [29, 58, 59], social media understanding [1, 16, 49], healthcare
20 interaction [8, 14, 33], and affect-aware robotics [10, 11, 36]. Unlike unimodal sentiment analysis,
21 MSA can use what a person says, how they say it, and how they look while saying it. This
22 makes the task closer to sentiment understanding in everyday communication [30]. Prior work has
23 improved MSA through multimodal representation learning [9, 22, 51], cross-modal alignment [32,
24 56], modality fusion [23, 35, 54], and modeling modality consistency, conflict, noise, and bias [20,
25 23, 45, 51, 56].

26 Recent multimodal large language models (MLLMs) have become strong general-purpose multimodal
27 learners [3, 44], with growing ability in multimodal reasoning [13, 42]. This has led researchers
28 to test MLLMs on emotion-related tasks [18], use them to add contextual or emotion-description
29 information to MSA models [40, 41], and transfer their knowledge to task-specific models through
30 lightweight adapters or MLLM-guided training [47, 57]. These studies show that MLLMs can work
31 well on MSA under standard benchmarks and aggregate metrics. What remains less clear is where
32 they fail. In particular, we still know little about whether they make stable decisions when two
33 affective states are close to each other.

34 We study this limitation in Section 3 by separating MSA into Coarse-grained Affective Recognition
35 (CAR) and Fine-grained Affective Recognition (FAR). CAR only checks whether the model gets
36 the sentiment direction right, while FAR requires a choice among ordered sentiment labels. On

CH-SIMS [50], CMU-MOSI [52], and CMU-MOSEI [2], MLLMs show a clear CAR–FAR gap: they often recognize polarity, but become much less reliable near fine-grained boundaries. Their FAR errors are mostly between adjacent labels. Layer-wise analysis shows the same pattern inside the model: wrong predictions often lack a stable gold-over-adjacent margin in middle-to-late layers. If we explicitly give the reference label and the adjacent competitor in an analysis-only setting, many of these margins come back (Appendix A). The boundary signal is there, but standard inference does not consistently bring it out.

We therefore propose **LSB-Distill** in Section 4. During training, the same MLLM runs in two modes. The student sees the normal multimodal input and predicts the FAR label. The teacher, used only for training, also receives the reference label and the strongest adjacent competitor selected by the current student. It then provides layer-wise targets for this local boundary. The student learns from both the FAR label and these latent boundary targets. During inference, we keep only the student. Experiments in Section 5 show that LSB-Distill improves FAR accuracy by **17.4** points and Macro-F1 by **22.0** points, while reducing Adjacent Error Rate by **12.9** points on average across representative MLLM backbones.

Our contributions are:

- We identify latent boundary formation as a central bottleneck of MLLMs in fine-grained multimodal sentiment analysis. Current MLLMs often recognize coarse sentiment polarity, but remain unreliable when distinguishing neighboring affective states in an ordered label space.
- We show that adjacent-label errors are linked to weak *gold-over-adjacent* separation in middle-to-late latent layers, revealing that fine-grained affective failures arise from unstable internal boundary formation rather than only final-layer misclassification.
- We propose LSB-Distill, a training-only boundary distillation method that transfers reference-conditioned local boundary trajectories to the standard student mode, improving boundary separation without any additional inference-time cost.
- Experiments on CH-SIMS, CMU-MOSI, and CMU-MOSEI show that LSB-Distill improves FAR accuracy and Macro-F1 while reducing adjacent-boundary errors across representative MLLM backbones, with ablations validating the role of local latent boundary supervision.

2 Related Work

MSA studies how language, speech, and visual behavior jointly express sentiment. Earlier task-specific methods mainly improved multimodal fusion, from tensor fusion [19, 53] to memory-based sequential fusion [54]. Later work focused on cross-modal alignment and representation learning, including attention-based alignment [38], modality-invariant and modality-specific decomposition [9], self-supervised unimodal supervision [51], and language-guided multimodal learning [39, 56]. Recent studies further consider noisy or imbalanced modalities [23], modality bias [45], sentiment disagreement [20], causal confounding [12], and modality distribution gaps [21]. Other methods explore information-theoretic representations [22, 43], emotion descriptions [41], dynamic experts [6], and subspace disentanglement [25]. Different from these task-specific designs, our work studies how MLLMs form internal sentiment boundaries when they make fine-grained affective predictions.

Recent studies have further introduced LLMs and MLLMs into MSA. A growing body of work has examined how large models can support text-centric and multimodal sentiment analysis [46]. Some studies directly build or adapt large models for MSA, such as multimodal sentiment analysis large language models [55], LoRA-based MLLM fine-tuning [26], and MLLM-enhanced sentiment networks [60]. Other work uses large models as knowledge sources or representation enhancers by introducing contextual world knowledge, textual emotional descriptions, or hidden MLLM representations [40, 41, 60]. Recent studies also explore parameter-efficient adaptation and knowledge transfer, including mixture-of-multimodal adapters, lightweight multimodal plugins, and MLLM-guided training frameworks [4, 47, 57]. These works show that LLMs and MLLMs can improve MSA as predictors, knowledge sources, representation enhancers, or training supervisors. In contrast, our work does not aim to propose another MSA architecture; instead, we study the reliability of MLLM affective decisions and analyze how their errors are structured near adjacent sentiment-state boundaries.

3 Preliminary Study

To identify the core bottleneck of MLLMs in multimodal sentiment analysis, we conduct a preliminary study on three widely used benchmarks: CH-SIMS(v2) [50], CMU-MOSI [52], and CMU-MOSEI [2]. We evaluate seven representative models, including Emotion-LLaMA [5], Interactive-Omni [37], LLaVA-OneVision [17], MiniCPM-o 4.5 [48], Qwen2.5-Omni [44], GPT-4o mini [28], and GPT-4o [27]. To keep the fine-grained label space consistent across datasets, we map all annotations into a unified five-level ordered space: NEG < WNEG < NEU < WPOS < POS. For CMU-MOSI and CMU-MOSEI, the original HNEG and NEG labels are merged into NEG, while HPOS and POS are merged into POS. We evaluate two settings: coarse-grained affective recognition (CAR), which uses three sentiment-direction labels NEG, NEU, and POS, and fine-grained affective recognition (FAR), which uses the unified five-level ordered label space. We summarize the key findings as follows.

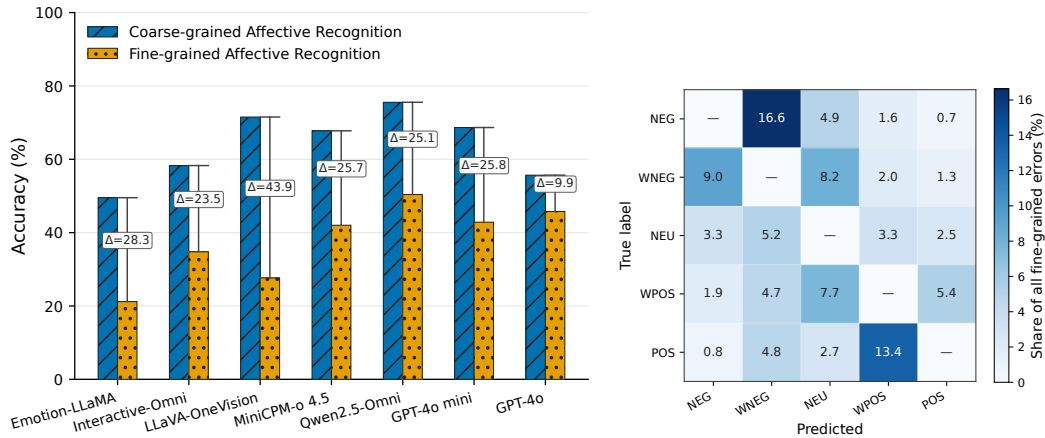


Figure 1: Preliminary analysis of MLLMs on MSA. Left: CAR and FAR performance of representative MLLMs on MSA benchmarks. Right: aggregated FAR error distribution across all models.

Finding I: FAR remains difficult for current MLLMs. As shown in Figure 1 (left), all models show limited absolute performance on FAR, indicating that fine-grained affective intensity recognition remains challenging. Most models also exhibit a clear CAR–FAR gap. For example, LLaVA-OneVision has a gap of 43.9 percentage points, while GPT-4o shows a much smaller gap of 9.9 percentage points. This suggests that current MLLMs can often capture the overall sentiment direction, but their ability to resolve fine-grained affective intensity varies substantially across models.

Finding II: FAR errors mainly occur near adjacent affective-state boundaries. As shown in Figure 1 (right), errors are not randomly distributed across all labels, but are concentrated between neighboring categories, such as NEG ↔ WNEG, WNEG ↔ NEU, NEU ↔ WPOS, and WPOS ↔ POS. This indicates that FAR failures are largely boundary confusions in an ordered affective space, rather than ordinary classification errors.

Finding III: Adjacent errors arise from failed boundary advantage formation. We compare Correct-FAR and Adjacent-Error samples by scoring all FAR candidates and focusing on the local competition between the gold label y and its strongest adjacent competitor a . The output-level results show a clear gap: in dataset-level and model-level summaries, Adjacent-Error samples have higher gold rank (2.34/2.32 vs. 1.12/1.44), lower boundary score margin $m_{bd} = s(y) - s(a)$ (−1.04/ −0.63 vs. +2.15/ +2.33), lower signed boundary margin $\Delta p_{bd} = p(y) - p(a)$ (−0.32/ −0.13 vs. +0.56/ +0.47), and higher boundary-local entropy (0.83/0.68 vs. 0.43/0.48). Thus, the gold label often remains plausible, but it fails to beat its adjacent rival. This failure also appears before final decoding. On Qwen2.5-Omni with CH-SIMS, Correct-FAR samples gradually build positive gold-over-adjacent margins in middle-to-late layers (−0.54 → +5.15 → +7.59 → +2.41), while Adjacent-Error samples stay negative (−10.90 → −15.10 → −13.64 → −1.59), even with some late-layer recovery. Reference-conditioned boundary analysis further shows that explicitly giving the reference label and adjacent competitor can turn ambiguous or negative margins into positive ones. This suggests that MLLMs contain useful boundary-separation signals, but standard inference does not reliably activate them. More results are in Appendix A. This observation motivates LSB-Distill, introduced in Section 4.

4 Method

4.1 Overview of LSB-Distill

The preliminary study in Section 3 shows that adjacent FAR errors are tied to a specific internal failure: the model often does not build a stable margin between the gold label and its strongest neighboring label in middle-to-late layers. Based on this finding, we propose LSB-Distill, a latent boundary distillation method for fine-grained multimodal sentiment analysis. The goal is simple: besides learning the correct FAR label, the model should also learn to separate that label from the adjacent category it is most likely to confuse with.

Figure 2 gives the pipeline. For each training sample, the student first uses the normal inference prompt and scores all FAR labels. Among the labels adjacent to the gold label, we pick the one with the highest student score as the strongest competitor. This gives a sample-specific local boundary. A teacher pass then takes the same multimodal input, together with the gold label and this competitor. The teacher does not generate explanations. We only use its layer-wise scores on the two boundary labels as training targets. The student is trained with both standard FAR supervision and this latent boundary signal, so that the gold label gains an advantage over its adjacent competitor before final decoding. The teacher is used only during training. At test time, the model receives the standard multimodal input, uses the student mode only, and predicts the FAR label in the same way as a normal candidate-scoring model.

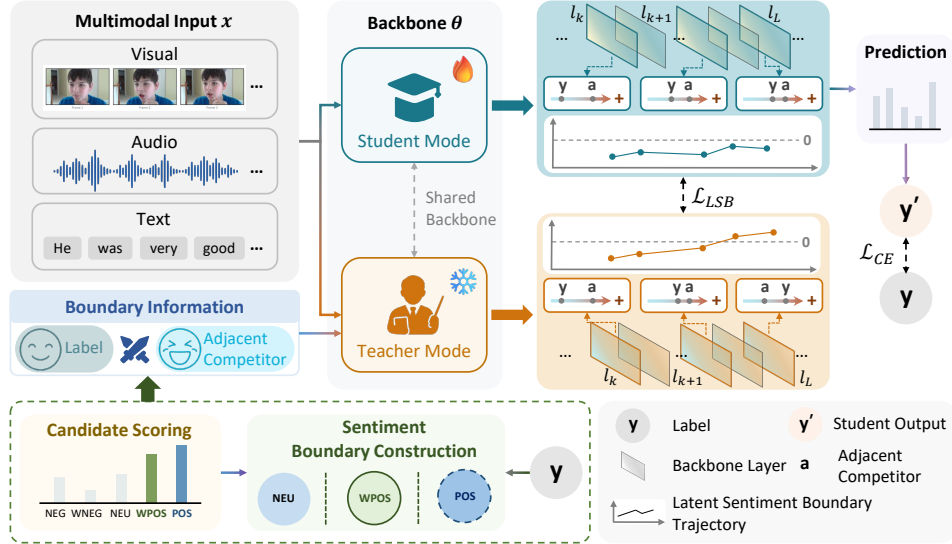


Figure 2: Overview of LSB-Distill. The student predicts the FAR label from the standard multimodal input. During training, the teacher also sees the gold label and its strongest adjacent competitor, and its layer-wise boundary signals are used to guide the student.

4.2 Layer-wise Latent Candidate Scoring

We first describe how FAR candidate scores are obtained at both the final layer and intermediate latent layers. Let $\mathcal{Y}$ denote the ordered FAR label set, including NEG, WNEG, NEU, WPOS, and POS. Instead of relying on free-form generation, we evaluate each candidate label by its conditional likelihood under the same prompt. For a candidate label $c \in \mathcal{Y}$, let its tokenized form be $\mathcal{T}(c) = \{t_1^c, \dots, t_{R_c}^c\}$, where R_c is the number of label tokens. Given an input prompt x_i^M , where $M \in \{S, T\}$ denotes student mode or teacher mode, the final-layer candidate score is defined as:

$$s_M^{(L)}(c) = \frac{1}{R_c} \sum_{r=1}^{R_c} \log p_{\theta}(t_r^c | x_i^M, t_{<r}^c), \quad (1)$$

where L denotes the final layer. The length normalization reduces the bias caused by different tokenization lengths. In student mode, the final FAR prediction is obtained by selecting the candidate with the highest final-layer score.

156 To analyze and supervise the internal boundary formation process, we further reconstruct candidate
 157 scores from intermediate layers. When scoring candidate c , let $h_{M,i,r-1}^{(l,c)}$ denote the hidden state at
 158 layer l before predicting the candidate token t_r^c under mode M . We project this hidden state to the
 159 vocabulary space using the original language-model head:

$$\ell_{M,i,r}^{(l,c)} = W_{\text{lm}} \text{LN} \left(h_{M,i,r-1}^{(l,c)} \right) + b_{\text{lm}}, \quad (2)$$

160 where W_{lm} , b_{lm} , and $\text{LN}(\cdot)$ denote the language-model head and the final normalization. The
 161 layer-wise score of candidate c is then computed as:

$$s_M^{(l)}(c) = \frac{1}{R_c} \sum_{r=1}^{R_c} \log \text{softmax} \left(\ell_{M,i,r}^{(l,c)} \right)_{t_r^c}. \quad (3)$$

162 This layer-wise candidate scoring can be viewed as a logit-lens style reconstruction of the model’s
 163 evolving preference over FAR labels. These scores provide the latent states used for sentiment
 164 boundary construction and distillation.

165 4.3 Student-Adaptive Sentiment Boundary Construction

166 FAR labels form an ordered sentiment space rather than independent categories. Following the
 167 ordered label space, we define the ordinal index function $o(\cdot)$ by mapping NEG, WNEG, NEU, WPOS, and
 168 POS to 0, 1, 2, 3, 4, respectively. For each gold label y_i , its adjacent label set is defined as:

$$\mathcal{N}(y_i) = \{y \in \mathcal{Y} : |o(y) - o(y_i)| = 1\}. \quad (4)$$

169 For example, if $y_i = \text{WPOS}$, its adjacent labels are NEU and POS. Since our preliminary analysis shows
 170 that FAR errors mainly occur near adjacent boundaries, LSB-Distill focuses on the local boundary
 171 between the gold label and its most competitive adjacent label.

172 Specifically, for each training sample, we select the strongest adjacent competitor according to the
 173 current student scores:

$$a_i = \arg \max_{a \in \mathcal{N}(y_i)} s_S^{(L)}(a), \quad (5)$$

174 where $s_S^{(L)}(\cdot)$ denotes the student-mode final-layer candidate score defined in Eq. 1. The pair (y_i, a_i)
 175 defines a sample-specific sentiment boundary. This construction is adaptive to the current student:
 176 the gold label y_i is provided by supervised training data, while the competing label a_i is selected
 177 according to the student’s current belief and may change as training progresses.

178 At layer l , the student’s gold-over-adjacent boundary margin is defined as:

$$m_{S,i}^{(l)} = s_S^{(l)}(y_i) - s_S^{(l)}(a_i). \quad (6)$$

179 A positive value of $m_{S,i}^{(l)}$ means that the gold label dominates its adjacent competitor at layer l , while
 180 a negative value means that the adjacent competitor is locally preferred. In the basic version of
 181 LSB-Distill, all training samples are used; we do not introduce an additional hard-sample threshold.
 182 Instead, each sample contributes a student-adaptive local boundary defined by Eq. 5. This keeps the
 183 training objective simple while preserving the key on-policy property that the boundary is selected
 184 according to the current student.

185 4.4 Latent Sentiment Boundary Distillation

186 **Reference-conditioned teacher mode.** During supervised training, the reference label y_i is avail-
 187 able, as in standard cross-entropy learning. LSB-Distill uses this information to construct a reference-
 188 conditioned teacher mode that estimates how the model represents the specified local sentiment
 189 boundary when the relevant boundary is made explicit. The student mode input is the standard
 190 multimodal input:

$$x_i^S = x_i,$$

191 while the teacher mode input is:

$$x_i^T = [x_i; y_i; a_i; \text{boundary query}], \quad (7)$$

where a_i is the strongest adjacent competitor selected by the current student. The teacher mode shares the same model backbone but is evaluated under the reference-conditioned prompt. Its outputs are detached and used only as training targets. Importantly, the teacher mode is not a valid inference setting and is never used during validation or testing.

For each selected middle-to-late layer $l \in \mathcal{L}_{mid}$, we construct the student boundary-local distribution over the pair (y_i, a_i) :

$$q_{S,i}^{(l)} = \text{softmax} \left(\frac{[s_S^{(l)}(y_i), s_S^{(l)}(a_i)]}{\tau} \right), \quad (8)$$

and the corresponding teacher boundary-local distribution:

$$q_{T,i}^{(l)} = \text{softmax} \left(\frac{[s_T^{(l)}(y_i), s_T^{(l)}(a_i)]}{\tau} \right), \quad (9)$$

where τ is the temperature. The sequence of boundary-local distributions over $\mathcal{L}_{mid}$ forms a latent sentiment boundary trajectory, describing how the model separates the gold label from its adjacent competitor across layers.

Learning objective. We distill the teacher’s latent sentiment boundary trajectory into the student using teacher-to-student boundary cross-entropy:

$$\mathcal{L}_{LSB} = -\frac{1}{N} \sum_{i=1}^N \sum_{l \in \mathcal{L}_{mid}} \omega_l \sum_{u \in \{y_i, a_i\}} \text{stopgrad} \left(q_{T,i}^{(l)}(u) \right) \log q_{S,i}^{(l)}(u), \quad (10)$$

where N is the number of training samples, ω_l is the layer weight, and $\sum_{l \in \mathcal{L}_{mid}} \omega_l = 1$. This loss is equivalent to optimizing the student-dependent term of $D_{KL}(q_{T,i}^{(l)} \| q_{S,i}^{(l)})$. Since the distribution is defined only over $\{y_i, a_i\}$, Eq. 10 directly supervises the local sentiment boundary between the gold label and the strongest adjacent competitor, rather than distilling the full five-label distribution.

In addition, we apply standard FAR supervision at the final layer. The student final-layer candidate distribution is:

$$p_S^{(L)}(c) = \frac{\exp(s_S^{(L)}(c))}{\sum_{c' \in \mathcal{Y}} \exp(s_S^{(L)}(c'))}, \quad (11)$$

and the classification loss is defined as:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \log p_S^{(L)}(y_i). \quad (12)$$

Overall, the learning objective of LSB-Distill is:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{LSB}, \quad (13)$$

where λ controls the strength of latent sentiment boundary distillation.

During inference, LSB-Distill introduces no additional teacher computation. The model receives only the standard multimodal input, scores FAR candidates under the student mode, and predicts:

$$\hat{y}_i = \arg \max_{c \in \mathcal{Y}} s_S^{(L)}(c). \quad (14)$$

Thus, the reference-conditioned teacher mode is only a training-time mechanism for shaping latent sentiment boundaries, while the inference procedure remains the same as standard candidate-based FAR prediction.

5 Experiment

5.1 Experimental Setup

Datasets. We use three standard MSA benchmarks: CH-SIMS [50] (v2), CMU-MOSI [52], and CMU-MOSEI [2]. CH-SIMS is a Chinese dataset built from movies, TV series, and variety shows. We use its v2 version in our experiments. It provides sentiment labels for both multimodal inputs and single modalities. CMU-MOSI contains English opinion videos from YouTube, with utterance-level sentiment intensity annotations. CMU-MOSEI follows a similar setting but is much larger, with more speakers, topics, and sentiment expressions.

Table 1: Performance comparison. FAR and F1 are percentages; AER denotes Adjacent Error Rate. Best results are in bold and second-best results are underlined. [†] marks original-paper results; ⁺ marks original Acc-7 results used as stricter references; [‡] marks task-specific MSA models assisted by LLMs or MLLMs.

Method	CH-SIMS			CMU-MOSI			CMU-MOSEI		
	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$
<i>Task-specific MSA models</i>									
Self-MM [51]	44.02	39.57	38.39	28.35	27.53	50.47	37.00	19.17	39.70
C-MIB [22]	11.96	4.27	52.56	16.51	5.67	40.03	37.10	10.82	30.90
DLF [39]	43.32	40.00	34.97	24.61	19.98	44.70	37.10	10.84	30.80
DEVA [41]	44.12	37.33	41.91	31.00	28.01	49.53	38.10	28.61	45.70
EMOE [6]	41.71	36.44	39.60	28.82	27.27	42.52	35.90	20.14	37.10
MMSLF [57] [‡]	44.02	40.08	38.59	28.04	25.73	50.78	45.70	39.71	41.20
MSE-Adapter [47] [‡]	12.46	7.44	47.64	17.29	10.02	32.71	37.00	10.81	30.90
CaReFlow [21] [†]	57.90	–	–	50.60 ⁺	–	–	55.70 ⁺	–	–
TSD [25] [†]	–	–	–	49.00 ⁺	–	–	54.90 ⁺	–	–
<i>Zero-shot MLLM baselines</i>									
Emotion-LLaMA [5]	21.81	12.39	35.68	29.56	16.06	33.81	15.59	10.05	55.36
LLaVA-OneVision [17]	26.33	19.10	60.70	36.76	27.36	53.43	23.10	21.74	66.80
MiniCPM-o 4.5 [48]	40.80	30.39	45.33	44.39	36.03	46.88	55.31	38.21	29.05
Qwen2.5-Omni [44]	52.36	31.92	45.03	42.52	26.23	39.25	53.50	31.51	25.46
Interactive-Omni [37]	27.74	27.37	50.85	46.11	45.25	45.02	45.82	38.38	38.51
Gemma-4-E4B-it [7]	46.03	35.96	31.96	41.74	30.83	38.79	54.40	36.13	25.31
<i>MLLM backbone + LSB-Distill</i>									
Emotion-LLaMA + LSB-Distill	48.14	41.00	<u>29.25</u>	57.17	52.68	32.71	60.70	48.89	26.30
LLaVA-OneVision + LSB-Distill	47.84	47.21	40.30	61.99	59.08	31.00	54.70	<u>50.71</u>	34.10
MiniCPM-o 4.5 + LSB-Distill	58.69	55.59	26.03	<u>64.49</u>	<u>59.93</u>	<u>30.37</u>	63.26	53.32	20.00
Qwen2.5-Omni + LSB-Distill	58.35	52.14	31.79	64.72	62.03	27.85	62.19	50.64	22.30
Interactive-Omni + LSB-Distill	37.39	29.76	31.46	60.28	58.21	31.78	61.09	48.63	25.23
Gemma-4-E4B-it + LSB-Distill	52.46	48.90	33.67	52.18	48.92	34.58	51.70	42.81	26.80

Models and baselines. We evaluate LSB-Distill on several open-source MLLMs released between 2024 and 2026. These include Emotion-LLaMA [5], LLaVA-OneVision [17], MiniCPM-o 4.5 [48], Qwen2.5-Omni [44], Interactive-Omni [37], and Gemma-4-E4B-it [7]. The set covers emotion-oriented, vision-language, omni-modal, and Gemma-based multimodal backbones.

We also compare with task-specific MSA models, including Self-MM [51], C-MIB [22], DLF [39], DEVA [41], EMOE [6], and ITHP [43]. These methods cover several common lines of MSA modeling, such as self-supervised multimodal learning, information bottleneck learning, language-centered disentanglement, emotion-description fusion, and expert-based fusion. We further include recent MSA methods, CaReFlow [21] and TSD [25], as well as LLM-assisted MSA models, MMSLF [57] and MSE-Adapter [47].

Evaluation metrics. We report FAR accuracy and Macro-F1. Since FAR labels are ordered sentiment categories, we also report Adjacent Error Rate (AER), which counts errors that move the prediction to a neighboring label. Given the gold label y_i , prediction $\hat{y}_i$, and ordinal label index $o(\cdot)$, AER is

$$\text{AER} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i \neq y_i \wedge |o(\hat{y}_i) - o(y_i)| = 1].$$

Lower AER means fewer boundary errors between nearby sentiment states.

5.2 Main Results

Table 1 shows the main results. LSB-Distill brings clear gains for most MLLM backbones, especially on FAR, where the model must choose between nearby sentiment labels. On CH-SIMS, MiniCPM-o 4.5 + LSB-Distill gives the best MLLM result, with 58.69% FAR accuracy and 55.59% F1. On CMU-MOSI, Qwen2.5-Omni + LSB-Distill performs best, improving FAR accuracy from 42.52% to 64.72% and F1 from 26.23% to 62.03%. On CMU-MOSEI, MiniCPM-o 4.5 + LSB-Distill achieves the highest FAR accuracy and F1, while reducing AER from 29.05% to 25.46%. Compared with task-specific MSA models and recent reported results, our method is also competitive under the unified FAR setting. The main takeaway is that LSB-Distill does more than improve average accuracy: it helps the model handle the adjacent sentiment boundaries where fine-grained MSA is most likely to fail.

5.3 Ablation Study

Effect of LSB supervision. We next look at what the latent boundary loss contributes beyond standard FAR training. For each backbone, Backbone + LSB-Distill uses the full objective $\mathcal{L}_{CE} + \lambda\mathcal{L}_{LSB}$. We compare it with *w/o LSB*, which keeps only candidate-level FAR supervision, and *w/o CE*, which keeps only latent boundary distillation. As shown in Table 2, the full objective gives the strongest results across backbones and datasets. Removing CE drops FAR by 3.66 points and Macro-F1 by 4.97 points on average, while AER increases by 4.34 points. Removing LSB causes a larger decline: FAR drops by 11.40 points, Macro-F1 drops by 17.04 points, and AER rises by 8.69 points. These results suggest a simple division of labor. CE keeps the prediction tied to the correct FAR label, while LSB gives extra pressure on the local boundary between neighboring sentiment categories.

Table 2: Ablation study on latent boundary supervision across MLLM backbones and datasets.

Method	CH-SIMS			CMU-MOSI			CMU-MOSEI		
	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$	FAR $\uparrow$	Macro-F1 $\uparrow$	AER $\downarrow$
Emotion-LLaMA + LSB-Distill	48.14	41.00	29.25	57.17	52.68	32.71	60.70	48.89	26.30
<i>w/o CE</i>	41.01	34.83	33.84	51.20	46.00	36.80	54.80	42.50	28.90
<i>w/o LSB</i>	36.50	25.80	34.90	39.80	30.50	41.50	49.20	32.80	31.60
LLaVA-OneVision + LSB-Distill	47.84	47.21	40.30	61.99	59.08	31.00	54.70	50.71	34.10
<i>w/o CE</i>	45.88	41.28	41.36	58.70	54.20	34.50	50.80	45.60	37.20
<i>w/o LSB</i>	42.50	34.00	43.50	43.80	36.50	46.80	39.20	31.80	48.50
MiniCPM-o 4.5 + LSB-Distill	58.69	55.59	26.03	64.49	59.93	30.37	63.26	53.32	20.00
<i>w/o CE</i>	57.69	45.43	36.83	60.80	54.40	34.20	59.10	47.50	29.80
<i>w/o LSB</i>	51.40	38.20	39.50	45.90	35.60	40.50	52.30	36.80	32.70
Qwen2.5-Omni + LSB-Distill	58.35	52.14	31.79	64.72	62.03	27.85	62.19	50.64	22.30
<i>w/o CE</i>	56.64	50.83	35.61	62.62	60.02	30.76	60.89	45.42	28.04
<i>w/o LSB</i>	55.94	39.61	37.16	44.16	29.85	38.96	55.36	35.27	30.31
Interactive-Omni + LSB-Distill	37.39	29.76	45.03	60.28	58.21	31.78	61.09	48.63	25.23
<i>w/o CE</i>	34.20	28.00	47.30	56.23	55.16	36.60	56.40	43.70	29.60
<i>w/o LSB</i>	31.80	22.60	48.80	46.88	47.08	43.30	49.50	34.60	33.80
Gemma-4-E4B-it + LSB-Distill	52.46	48.90	33.67	52.18	48.92	34.58	51.70	42.81	26.80
<i>w/o CE</i>	47.54	44.04	38.29	48.70	43.50	37.20	48.30	38.60	30.40
<i>w/o LSB</i>	43.20	32.50	39.70	39.60	28.90	42.30	45.10	31.40	31.70

Effect of LSB-Distill internal modules. We further check the main design choices inside LSB-Distill. For each dataset, we use a strong backbone from the main comparison and replace one module at a time, while keeping the rest of the pipeline fixed. As shown in Table 3, the full design gives the best overall results. The hard boundary target hurts most, with average drops of 10.16 points in FAR and 13.83 points in F1. This indicates that the reference-conditioned teacher provides useful soft boundary signals, rather than simply pushing the model toward a hard gold label. Full-label distillation is also weaker than the local gold-vs-adjacent objective. The layer range matters too. Final-layer-only, early-layer-only, and all-layer variants all perform worse than the middle-to-late trajectory; on CMU-MOSI, final-layer-only distillation reduces FAR from 64.72% to 61.40%. Using all adjacent labels or a random adjacent label is also less effective than selecting the strongest adjacent competitor, which shows that the boundary should be tied to the student’s own current confusion.

Hyperparameter sensitivity. We test the sensitivity of LSB-Distill to the LSB weight λ and temperature τ on Qwen2.5-Omni / CH-SIMS, with full results in Appendix C.2. The λ sweep shows that stronger boundary supervision is not always better. A moderate value, $\lambda = 1$, gives the highest FAR and Macro-F1, while $\lambda = 10$ gives the lowest AER but noticeably hurts Macro-F1. Temperature shows a similar pattern. The best result comes from $\tau = 1.5$, indicating that a slightly smoother teacher target works better than an overly sharp boundary target for adjacent sentiment categories.

5.4 Boundary Error and Latent Formation Analysis

Figure 3 summarizes how LSB-Distill changes boundary-level errors. Averaged over the six open-source MLLM backbones, AER drops from 44.93% to 32.08% on CH-SIMS, from 42.86% to 31.38% on CMU-MOSI, and from 40.08% to 25.79% on CMU-MOSEI. The reduction is not limited

Table 3: Ablation of LSB-Distill modules on representative backbone-dataset pairs. **Ours** is the full setting; each indented row changes one module.

Module / Variant	MiniCPM-o 4.5 CH-SIMS			Qwen2.5-Omni CMU-MOSI			Interactive-Omni CMU-MOSEI		
	FAR Acc ↑	F1 ↑	AER ↓	FAR Acc ↑	F1 ↑	AER ↓	FAR Acc ↑	F1 ↑	AER ↓
<i>Teacher target</i>									
Ours: reference-conditioned	58.69	55.59	26.03	64.72	62.03	27.85	61.09	48.63	25.23
<i>Hard boundary target</i>	50.59	45.11	30.95	50.47	38.26	35.28	52.95	41.40	31.65
<i>Gold-only teacher</i>	50.10	45.10	31.46	60.35	55.80	31.20	56.78	45.55	28.47
<i>Boundary scope</i>									
Ours: local boundary	58.69	55.59	26.03	64.72	62.03	27.85	61.09	48.63	25.23
<i>Full-label distillation</i>	54.97	49.55	32.36	59.80	54.60	32.10	56.80	43.70	29.90
<i>Layer scope</i>									
Ours: middle-to-late	58.69	55.59	26.03	64.72	62.03	27.85	61.09	48.63	25.23
<i>Final-layer only</i>	54.79	54.97	31.26	61.40	57.20	30.40	57.60	44.50	29.20
<i>Early layers only</i>	50.04	54.07	32.46	62.97	56.05	28.16	55.40	42.10	31.40
<i>All layers</i>	57.79	44.95	27.44	60.90	55.70	31.60	58.20	44.00	28.70
<i>Boundary construction</i>									
Ours: strongest adjacent	58.69	55.59	26.03	64.72	62.03	27.85	61.09	48.63	25.23
<i>All adjacent labels</i>	56.70	52.86	28.64	61.75	58.10	30.05	58.90	46.20	27.80
<i>Random adjacent label</i>	50.99	47.65	32.46	58.90	54.20	33.25	55.70	42.80	31.10

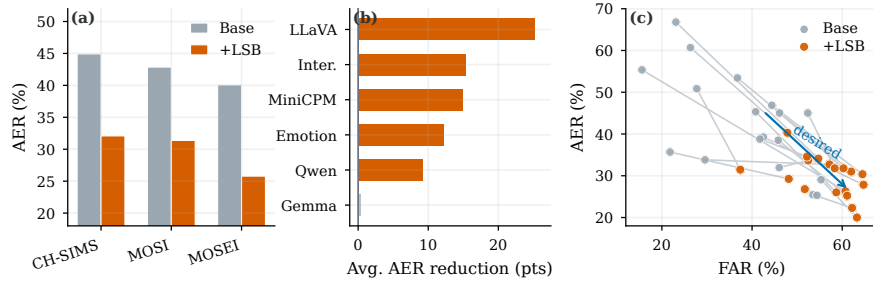


Figure 3: Boundary error analysis. Left: average AER before and after LSB-Distill across datasets. Middle: average AER reduction for each MLLM backbone. Right: FAR–AER movement for all backbone–dataset pairs, where each line links the zero-shot backbone to its LSB-Distill counterpart.

to one architecture: most backbones move toward lower AER, with the largest average drop observed on LLaVA-OneVision. The FAR–AER scatter gives the same picture from another angle. After LSB-Distill, most backbone–dataset points move toward higher FAR and lower AER, meaning that the gain is not just higher top-line accuracy but fewer mistakes across neighboring sentiment states. This also agrees with the latent analysis in Appendix A: Correct-FAR samples tend to form positive gold-over-adjacent margins in middle-to-late layers, whereas Adjacent-Error samples often remain weak or negative. Reference-conditioned probes further show that these local boundaries can be recovered when the relevant gold-adjacent pair is made explicit, which explains why the LSB teacher signal helps the student build more reliable sentiment boundaries.

6 Conclusion

This paper studied how current MLLMs behave in fine-grained multimodal sentiment analysis. Across CH-SIMS, CMU-MOSI, and CMU-MOSEI, we found that these models often capture coarse sentiment polarity, but their predictions become less reliable near neighboring affective states. The layer-wise analysis gives a clearer explanation: many adjacent-category errors occur when the model fails to build a stable gold-over-adjacent advantage in its middle-to-late latent layers. Based on this observation, we proposed LSB-Distill, which uses a training-only teacher mode to provide layer-wise supervision for local sentiment boundaries and transfers this signal to the standard student mode. Experiments on several MLLM backbones show that LSB-Distill improves FAR performance and reduces adjacent-category errors. The ablation results further show that final-label supervision and latent boundary supervision are complementary: one anchors the global label decision, while the other helps separate nearby sentiment states. These findings suggest that directly modeling latent sentiment boundaries can make MLLMs more reliable for fine-grained MSA.

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A Additional Details of the Preliminary Study

Purpose. The preliminary study is designed to locate the main bottleneck of current MLLMs in multimodal sentiment analysis. Instead of only reporting aggregate performance, we separate the evaluation into coarse-grained affective recognition (CAR) and fine-grained affective recognition (FAR). This allows us to examine whether a model that captures the overall sentiment direction can also make stable decisions near adjacent affective-state boundaries.

Datasets and models. We conduct the study on the official test splits of three widely used MSA benchmarks: CH-SIMS [50], CMU-MOSI [52], and CMU-MOSEI [2]. We evaluate seven representative MLLMs: Emotion-LLaMA [5], Interactive-Omni [37], LLaVA-OneVision [17], MiniCPM-o 4.5 [48], Qwen2.5-Omni [44], GPT-4o mini [28], and GPT-4o [27]. For models that support audio input, we provide the available audio-visual-textual inputs. For vision-language models without native audio support, we provide visual frames and the transcript. All models are evaluated in a zero-shot setting without task-specific fine-tuning.

Task settings. We evaluate two settings. CAR predicts three sentiment-direction labels: NEG, NEU, and POS. FAR predicts five ordered intensity labels: NEG, WNEG, NEU, WPOS, and POS, where WNEG and WPOS denote weak negative and weak positive sentiment. The label order is defined as

$$\text{NEG} < \text{WNEG} < \text{NEU} < \text{WPOS} < \text{POS}.$$

We report accuracy for both CAR and FAR, and compute the CAR–FAR gap as

$$\Delta = \text{Acc}_{\text{CAR}} - \text{Acc}_{\text{FAR}}.$$

For FAR, we further analyze the confusion matrix over the ordered label space. We define an adjacent-category error as an error where the predicted label and the gold label are neighbors in the above order.

To ensure the comparability of baselines, we use deterministic generation for all models, with temperature set to 0.0, top-p set to 1.0, and max new tokens set to 6. All open-source models are evaluated on NVIDIA A800 80GB GPUs with 8 CPU cores and 64GB RAM. For API-based models, we use the same prompts, label space, and deterministic decoding configuration whenever supported.

Prompt format. We use separate prompts for CAR and FAR. The CAR prompt asks the model to predict the overall sentiment direction:

You are given a short video segment, its audio if available, and the transcript. Transcript: {transcript}. Please infer the speaker’s overall sentiment. Choose exactly one label from: NEG, NEU, POS. NEG means negative sentiment, NEU means neutral sentiment, and POS means positive sentiment. Return only the label.

The FAR prompt asks the model to distinguish adjacent affective intensity levels:

You are given a short video segment, its audio if available, and the transcript. Transcript: {transcript}. Please infer the speaker’s fine-grained sentiment intensity. Choose exactly one label from: NEG, WNEG, NEU, WPOS, POS. NEG means clearly negative, WNEG means weakly negative, NEU means neutral, WPOS means weakly positive, and POS means clearly positive. Return only the label.

We parse the model output by exact label matching. If multiple valid labels appear, we use the first valid label. If no valid label is found, the prediction is treated as incorrect.

Detailed CAR–FAR results. Figure 4 provides a more detailed view of the CAR–FAR comparison, including the overall result and the results on each dataset. The gap between CAR and FAR is consistently observed across CH-SIMS, CMU-MOSEI, and CMU-MOSI, suggesting that the difficulty of FAR is not caused by a single dataset or annotation scheme. Across the three datasets, most models obtain substantially higher accuracy on CAR than on FAR. This indicates that current MLLMs are often able to identify the coarse sentiment direction, but still struggle to make stable decisions when the label space is refined into adjacent intensity levels.

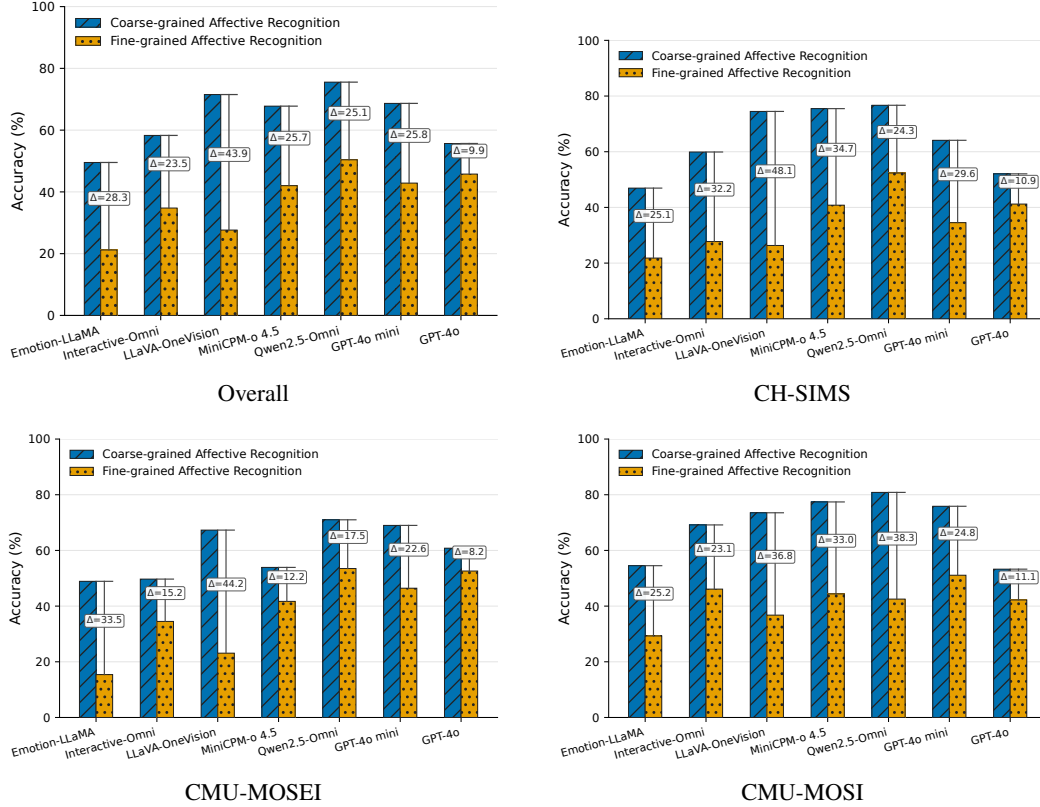


Figure 4: Detailed CAR-FAR performance of representative MLLMs. Overall results and dataset-level results show consistent performance drops from coarse-grained affective recognition to fine-grained affective recognition. The gap Δ denotes the difference between CAR accuracy and FAR accuracy.

596 The magnitude of the gap also varies clearly across models. LLaVA-OneVision shows the largest
 597 CAR-FAR gaps on all three datasets, with $\Delta = 48.1$ on CH-SIMS, $\Delta = 44.2$ on CMU-MOSEI, and
 598 $\Delta = 36.8$ on CMU-MOSI. In contrast, GPT-4o shows much smaller gaps, with $\Delta = 10.9$, $\Delta = 8.2$,
 599 and $\Delta = 11.1$ on the three datasets, respectively. This suggests that different MLLMs differ not only
 600 in their overall recognition accuracy, but also in their ability to preserve performance when moving
 601 from coarse sentiment direction to fine-grained affective intensity. Moreover, several models with
 602 strong CAR performance still exhibit large FAR drops, indicating that high coarse-grained accuracy
 603 does not necessarily imply calibrated fine-grained boundary decisions.

604 **Per-model FAR error structure.** Figure 5 further reports the per-model FAR error matrices. Each
 605 row is normalized by the number of samples from the corresponding true sentiment label. Diagonal
 606 entries are omitted, so the heatmap focuses on where each true label is misclassified rather than how
 607 often it is correctly predicted. Therefore, darker cells indicate the dominant error destinations for a
 608 given true label.

609 The results show that FAR errors are not uniformly distributed across the label space. For most
 610 models, many errors occur between adjacent or near-adjacent sentiment states, such as NEG $\rightarrow$ WNEG,
 611 WNEG $\rightarrow$ NEU, NEU $\rightarrow$ WPOS, and WPOS $\rightarrow$ POS. This pattern supports our main observation that FAR
 612 failures are largely boundary confusions in an ordered affective space. In other words, the model
 613 often fails at deciding whether the sentiment intensity has crossed a neighboring boundary, rather
 614 than confusing arbitrary labels.

615 The heatmaps also reveal model-specific error biases. For example, some models show strong
 616 attraction toward weak or neutral labels, indicating conservative intensity estimation, while others
 617 over-predict stronger positive or negative labels. These biases suggest that different MLLMs do
 618 not fail in the same way: some suffer from boundary ambiguity around neutral sentiment, whereas
 619 others exhibit systematic calibration shifts toward specific sentiment regions. Together, these results

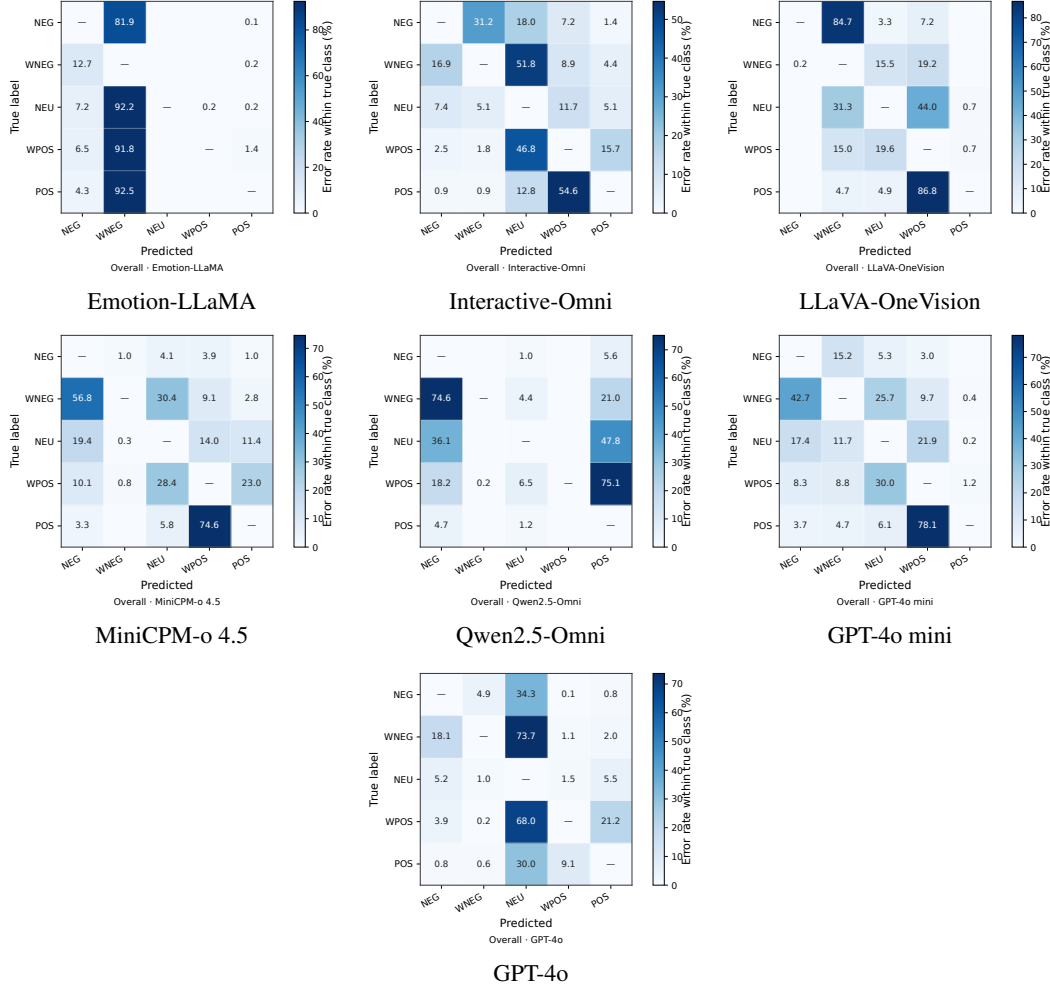


Figure 5: Per-model FAR error distributions. The confusion patterns show that many errors occur between adjacent affective categories, indicating that fine-grained failures are mainly boundary confusions rather than random misclassifications.

620 motivate a boundary-sensitive correction mechanism that explicitly contrasts the gold label with its
621 adjacent confounding labels.

622 **Mechanism analysis of boundary advantage formation.** Figure 3 gives a compact summary in
623 the main text. This appendix expands the analysis with metric definitions, per-model error heatmaps,
624 dataset-level and model-level layer curves, reference-conditioned probes, and qualitative cases.

625 To understand why FAR fails near adjacent affective boundaries, we conduct a candidate-level and
626 layer-wise mechanism analysis. We focus on two groups of samples: Correct-FAR samples, where
627 the model predicts the fine-grained label correctly, and Adjacent-Error samples, where the prediction
628 falls into a neighboring category. For each sample, we score all FAR candidates and examine the
629 local competition between the gold label y and its strongest adjacent competitor

$$a = \arg \max_{a' \in \mathcal{N}(y)} s(a'),$$

630 where $\mathcal{N}(y)$ denotes the adjacent labels of y , and $s(\cdot)$ is the candidate score.

631 **Candidate-level diagnostics.** For each FAR candidate c , we compute its normalized conditional
632 score under the same input prompt:

$$s(c) = \frac{1}{|\mathcal{T}(c)|} \sum_{r=1}^{|\mathcal{T}(c)|} \log p_{\theta}(t_r^c \mid x, t_{<r}^c),$$

Table 4: Aggregated output-level boundary diagnostics. Dataset overall refers to Qwen2.5-Omni aggregated over CH-SIMS, CMU-MOSI, and CMU-MOSEI. Model overall refers to five open-source models aggregated on CH-SIMS. Adjacent-Error samples consistently show worse gold rank, lower boundary margins, and higher boundary-local entropy.

Aggregation	Group	Gold rank ↓	$m_{\text{bd}} \uparrow$	$\Delta p_{\text{bd}} \uparrow$	$H_{\text{bd}} \downarrow$	$d_{\text{bd}} \uparrow$
Dataset overall	Correct-FAR	1.12	+2.15	+0.56	0.43	2.21
Dataset overall	Adjacent-Error	2.34	−1.04	−0.32	0.83	1.15
Model overall	Correct-FAR	1.44	+2.33	+0.47	0.48	2.55
Model overall	Adjacent-Error	2.32	−0.63	−0.13	0.68	1.64

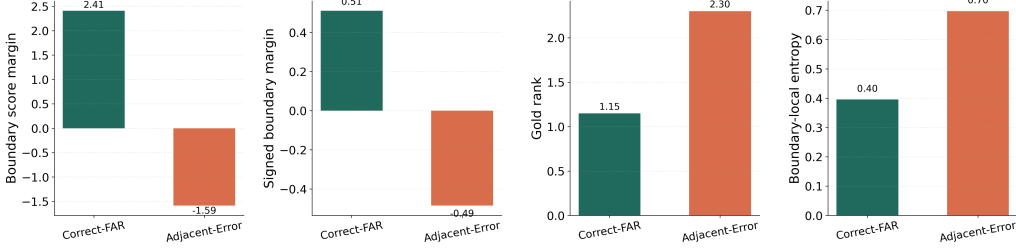


Figure 6: Output-level boundary diagnostics for Correct-FAR and Adjacent-Error samples. Adjacent-Error samples show higher gold rank, lower boundary margins, and higher boundary-local entropy, indicating that adjacent-category errors correspond to weak or failed local boundary advantage.

633 where $\mathcal{T}(c) = \{t_1^c, \dots, t_{|\mathcal{T}(c)|}^c\}$ denotes the tokenized candidate label. We normalize candidate
634 scores into probabilities:

$$p(c) = \frac{\exp(s(c))}{\sum_{c' \in \mathcal{Y}} \exp(s(c'))}.$$

635 We use five diagnostics. Gold rank measures the rank position of the gold label:

$$r_y = 1 + \sum_{c \in \mathcal{Y}, c \neq y} \mathbf{1}[s(c) > s(y)].$$

636 Boundary score margin measures the score gap between the gold label and its strongest adjacent
637 competitor:

$$m_{\text{bd}} = s(y) - s(a).$$

638 Signed boundary margin measures the same local competition in probability space:

$$\Delta p_{\text{bd}} = p(y) - p(a).$$

639 Boundary separation measures the magnitude of the local score gap:

$$d_{\text{bd}} = |s(y) - s(a)|.$$

640 Boundary-local entropy measures local uncertainty around the gold label:

$$H_{\text{bd}} = - \sum_{z \in \mathcal{B}(y)} q(z) \log q(z), \quad \mathcal{B}(y) = \{y\} \cup \mathcal{N}(y), \quad q(z) = \frac{p(z)}{\sum_{z' \in \mathcal{B}(y)} p(z')}.$$

641 Positive values of m_{bd} and Δp_{bd} indicate that the model has formed a gold-over-adjacent boundary
642 advantage.

643 Table 4 and Figure 6 show that Adjacent-Error samples have a higher gold rank, lower boundary score
644 margin, lower signed boundary margin, and higher boundary-local entropy under both dataset-level
645 and model-level aggregation. This suggests that the gold label remains locally plausible in many
646 adjacent-error cases, but it is not sufficiently separated from the strongest adjacent competitor.

647 **Layer-wise boundary advantage.** We further analyze when the boundary advantage forms inside
648 the MLLM. Since closed-source APIs do not expose intermediate states, this analysis is conducted on
649 open-source models with accessible hidden representations. For each selected layer l of the MLLM

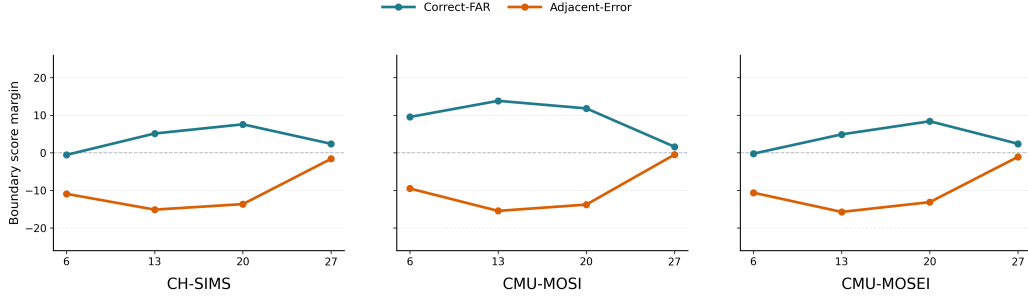


Figure 7: Dataset-level layer-wise boundary score margin analysis for Qwen2.5-Omni. Correct-FAR samples tend to form positive gold-over-adjacent margins in middle-to-late layers, whereas Adjacent-Error samples remain negative or only partially recover toward zero.

thinker, we reconstruct candidate-label scores from the hidden state and compute the layer-wise boundary score margin:

$$m_{\text{bd}}^{(l)} = s^{(l)}(y) - s^{(l)}(a),$$

and the layer-wise signed boundary margin:

$$\Delta p_{\text{bd}}^{(l)} = p^{(l)}(y) - p^{(l)}(a).$$

These quantities allow us to trace whether the model progressively forms a gold-over-adjacent advantage before final decoding.

Figure 7 shows dataset-level layer-wise curves for Qwen2.5-Omni. Across CH-SIMS, CMU-MOSI, and CMU-MOSEI, Correct-FAR samples generally form positive boundary margins in middle-to-late layers, while Adjacent-Error samples remain negative or only partially recover toward zero. For example, on CH-SIMS, Correct-FAR samples change from a slightly negative margin to a positive margin across layers:

$$-0.54 \rightarrow +5.15 \rightarrow +7.59 \rightarrow +2.41.$$

In contrast, Adjacent-Error samples remain negative:

$$-10.90 \rightarrow -15.10 \rightarrow -13.64 \rightarrow -1.59.$$

The final-layer recovery of Adjacent-Error samples suggests that the model still captures some boundary signal, but the signal is insufficient to reverse the adjacent competitor’s dominance.

Figure 8 further reports model-level layer-wise curves on CH-SIMS. Although different models exhibit different calibration behaviors, the aggregate pattern remains consistent: Correct-FAR samples form clearer boundary advantage than Adjacent-Error samples. Some Adjacent-Error curves move toward zero in late layers, but they are generally less stable and less consistently positive than Correct-FAR curves. This supports the view that adjacent-category errors arise from incomplete boundary advantage formation rather than arbitrary final-label mistakes.

Qualitative cases. To complement the group-level statistics, we provide qualitative cases from CH-SIMS, CMU-MOSEI, and CMU-MOSI. Each case study shows representative video frames, a transcript excerpt, the gold and predicted labels, candidate-label scores, and the layer-wise boundary score margin. These cases illustrate the same mechanism at the sample level: Correct-FAR examples develop a positive gold-over-adjacent margin, while Adjacent-Error examples either remain negative or recover only partially toward zero.

Reference-conditioned boundary analysis. The layer-wise analysis above shows that Adjacent-Error samples fail to form a positive gold-over-adjacent boundary margin under the standard input condition. We further ask whether this failure means that the model lacks the latent capacity to separate the boundary, or whether the boundary can be recovered when the relevant local decision is explicitly specified. To examine this, we conduct an answer-conditioned boundary probe. In this probe, the model receives the same multimodal input, but is additionally given the reference correct label y and its strongest adjacent competing label a . We then compute the same layer-wise boundary margin between y and a . This probe is used only for analysis and is not a valid inference setting.

The answer-conditioned boundary prompt is:

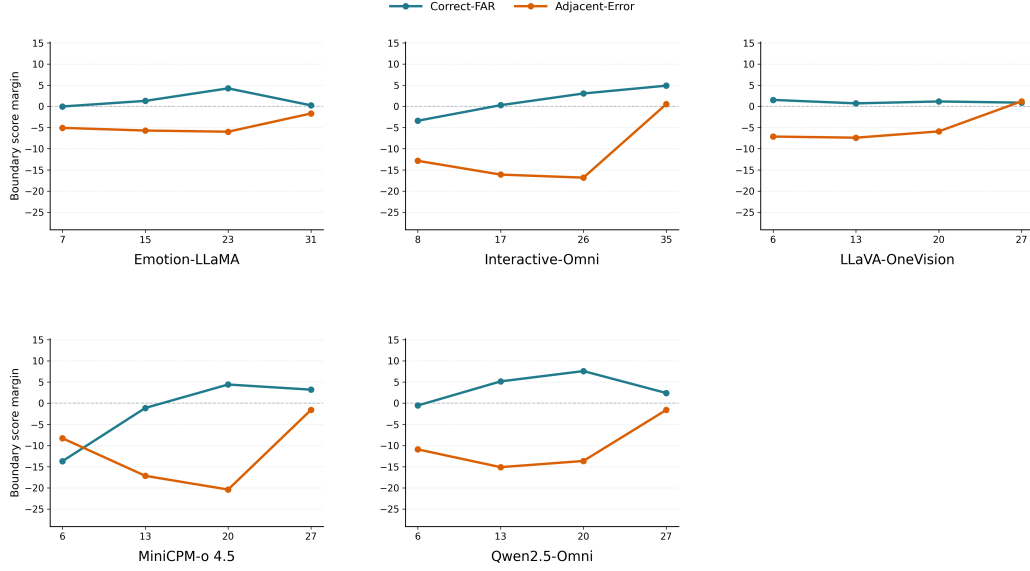


Figure 8: Model-level layer-wise boundary score margin analysis on CH-SIMS. Correct-FAR samples generally show stronger and more stable gold-over-adjacent margins than Adjacent-Error samples. The model-specific curves reveal that boundary formation failures vary across architectures, but Adjacent-Error samples consistently exhibit weaker local boundary advantage.

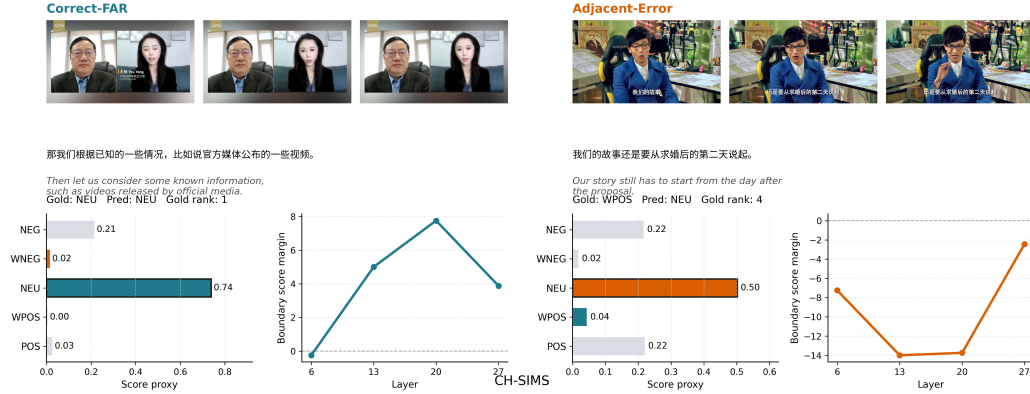


Figure 9: Qualitative boundary case study on CH-SIMS. The Correct-FAR case shows a dominant gold label and a positive layer-wise boundary margin. The Adjacent-Error case shows that the gold label fails to dominate its adjacent competitor, and the boundary margin remains negative despite partial late-layer recovery.

684 You are given a short video segment, its audio if available, and the transcript.
685 You are doing a local boundary probe for fine-grained multimodal sentiment recognition.
686 This probe is used only for analysis.
687 Reference correct label: [GOLD_LABEL]. Adjacent competing label:
688 [ADJACENT_COMPETITOR].
689 Compare only these two labels using the multimodal input. The reference correct label
690 should be preferred when evaluating this local boundary. Do not explain.
691 Transcript: [TRANSCRIPT]
692 Choose exactly one label from: [GOLD_LABEL], [ADJACENT_COMPETITOR]. Return only
693 the label.

694 As shown in Figure 12, answer conditioning substantially increases the gold-over-adjacent boundary
695 margin on adjacent-error cases. For Qwen2.5-Omni, the final-layer margin changes from -2.47 under
696 the standard prompt to $+1.13$ under the answer-conditioned prompt, with all three adjacent-error
697 cases flipping from negative to positive margins. For Interactive-Omni, the final-layer margin changes
698 from -2.44 to $+7.62$, and the mid-to-late layer average margin improves from $+0.91$ to $+6.30$.

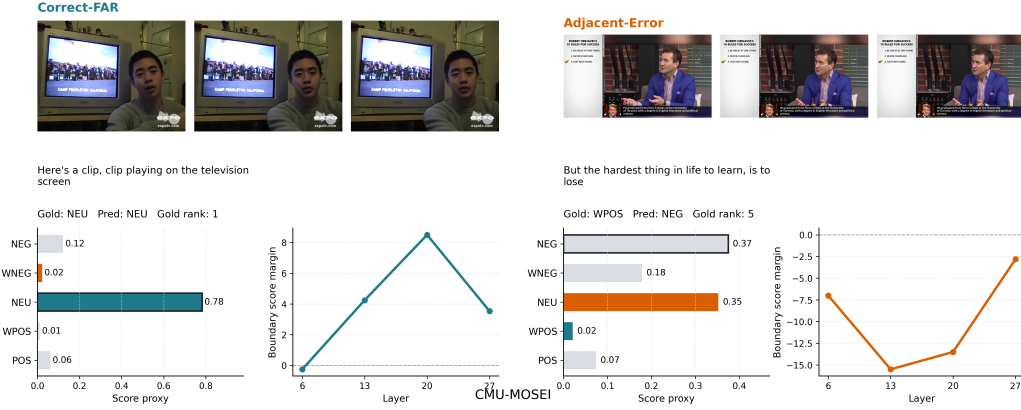


Figure 10: Qualitative boundary case study on CMU-MOSEI. The examples compare a Correct-FAR sample with an Adjacent-Error sample using video frames, transcript, candidate scores, and layer-wise boundary margins.



Figure 11: Qualitative boundary case study on CMU-MOSI. The Adjacent-Error case illustrates how the model can assign high scores to competing neighboring labels and fail to form a positive gold-over-adjacent margin across layers.

699 These results suggest that the model can represent the correct local sentiment boundary when the
700 relevant boundary is explicitly specified, but under the standard prompt it does not reliably form this
701 boundary advantage on its own.

702 B Supplement of Implementation Details

703 **Prompt templates.** We use two prompt modes during training. The student mode is the standard
704 multimodal sentiment recognition prompt and is also the only prompt used at inference time. The
705 teacher mode is a privileged boundary-scoring prompt used only for latent boundary distillation
706 during training.

707 Student prompt.

708 You are given a short video segment, its audio if available, and the transcript.

709 Transcript:
710 {transcript}

711 Please infer the speaker's fine-grained sentiment intensity.
712 Choose exactly one label from: NEG, WNEG, NEU, WPOS, POS.
713 NEG means clearly negative, WNEG means weakly negative, NEU means neutral,
714 WPOS means weakly positive, and POS means clearly positive.
715 Return only the label.

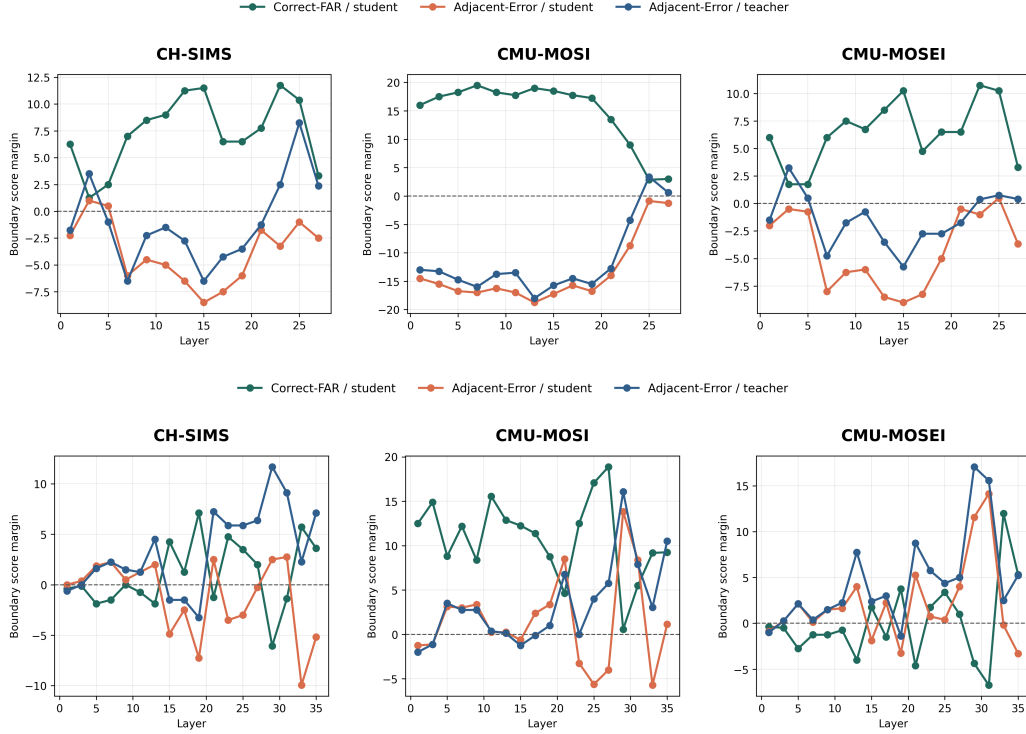


Figure 12: Answer-conditioned boundary probe. Top: Qwen2.5-Omni. Bottom: Interactive-Omni. The standard prompt uses only the multimodal input, while the answer-conditioned prompt additionally provides the reference correct label and its strongest adjacent competing label. On adjacent-error samples, answer conditioning converts ambiguous or negative gold-over-adjacent margins into positive margins, suggesting that the models have latent boundary-separation capacity but do not reliably activate it under the standard inference prompt.

718 **Teacher boundary prompt.**

719 You are a boundary teacher for fine-grained multimodal sentiment recognition.

720 The reference correct label is: {gold_label}.

721 The adjacent competing label from the current student is: {adjacent_competitor}.

722

723 Your task is to evaluate the local sentiment boundary between the reference

724 label and the adjacent competing label using the multimodal input.

725 The reference label should be preferred over the adjacent competing label

726 when computing the boundary target.

727 Do not generate explanations.

728

729 Transcript:

730 {transcript}

731

732 Choose one label from NEG, WNEG, NEU, WPOS, POS.

733 Answer:

734 For each training sample, the adjacent competitor is selected on-policy from the current student as

735 the highest-scoring adjacent label in the ordered label space.

736 **Label space.** We use two task settings in this work: coarse-grained affective recognition (CAR) and

737 fine-grained affective recognition (FAR). CAR evaluates whether the model can identify the overall

738 sentiment direction, and uses three labels: NEG, NEU, and POS. FAR evaluates whether the model can

739 distinguish finer sentiment states in an ordered label space. To make FAR comparable across datasets,

740 we use a unified five-level ordered label space:

$$\text{NEG} < \text{WNEG} < \text{NEU} < \text{WPOS} < \text{POS}.$$

Table 5: Dataset statistics and training hyperparameters for LSB-Distill.

Parameter	CH-SIMS	CMU-MOSI	CMU-MOSEI
Train split size	2717	1165	9520
Validation split size	646	193	1229
Test split size	1029	642	3750
Epochs	6	10	5
Batch size	4	4	4
Gradient accumulation	8	8	8
Optimizer	AdamW	AdamW	AdamW
Learning rate	2×10^{-5}	2×10^{-5}	2×10^{-5}
Weight decay	0	0	0
Gradient clipping	1.0	1.0	1.0
LSB weight λ	5.0	5.0	5.0
Temperature τ	1.0	1.0	1.0
Layer weights	uniform	uniform	uniform
LoRA rank	16	16	16
LoRA alpha	32	32	32
LoRA dropout	0.05	0.05	0.05

Table 6: Software environments used for LSB-Distill experiments.

Backbone	Python	PyTorch	Transformers	PEFT
Emotion-LLaMA [5]	3.10.20	2.0.0	4.35.0	0.2.0
LLaVA-OneVision [17]	3.11.15	2.5.1+cu121	4.52.0.dev0	0.19.1
MiniCPM-o 4.5 [48]	3.10.20	2.8.0	4.51.0	0.19.1
Qwen2.5-Omni [44]	3.11.15	2.5.1+cu121	4.52.0.dev0	0.19.1
Interactive-Omni [37]	3.11.15	2.5.1+cu121	4.52.0.dev0	0.19.1
Gemma-4-E4B-it [7]	3.11.15	2.11.0	5.5.4	0.19.1

Here, WNEG and WPOS denote weak negative and weak positive sentiment, respectively. CH-SIMS naturally follows this five-level setting. For CMU-MOSI and CMU-MOSEI, which are originally organized in a seven-level space, we merge HNEG and NEG into NEG, and merge HPOS and POS into POS. This conversion keeps all datasets in the same ordered FAR space and allows adjacent-category errors and sentiment boundaries to be defined consistently.

Hyperparameters. Table 5 summarizes the dataset statistics and training hyperparameters used for LSB-Distill. Before training, we remove samples with unavailable, corrupted, or unreadable video files, and report the final split sizes after this filtering step. We adjust the number of training epochs according to dataset size, with more epochs for the smaller CMU-MOSI split and fewer epochs for the larger CH-SIMS and CMU-MOSEI splits. Across datasets, we keep the optimization configuration consistent: all models are trained with AdamW, a batch size of 4, gradient accumulation of 8, learning rate 2×10^{-5} , no weight decay, and gradient clipping of 1.0. For the LSB-Distill objective, we use $\lambda = 5.0$, temperature $\tau = 1.0$, and uniform weights over the selected middle-to-late layers. We apply LoRA-based parameter-efficient tuning with rank 16, alpha 32, and dropout 0.05. During validation and testing, only the student mode is used, and the teacher boundary prompt is removed.

Computing environment. All experiments were conducted on NVIDIA A800-SXM4-80GB GPUs with 16 CPU cores and 160GB host memory. Each open-source MLLM experiment used one GPU worker per backbone–dataset setting, and ablation or hyperparameter runs used the same per-run allocation. Jobs were submitted with a 24-hour wall-clock limit and completed within this allocation; API-based baselines, when used, only required local CPU preprocessing. Since the evaluated MLLMs rely on different upstream codebases and dependency requirements, we used model-specific Python environments while keeping the same LSB-Distill training pipeline and configuration interface across backbones. All Python environments were managed with uv to ensure consistent dependency installation and reproducibility. The detailed software versions for each backbone, including Python, PyTorch, Transformers, and PEFT, are reported in Table 6.

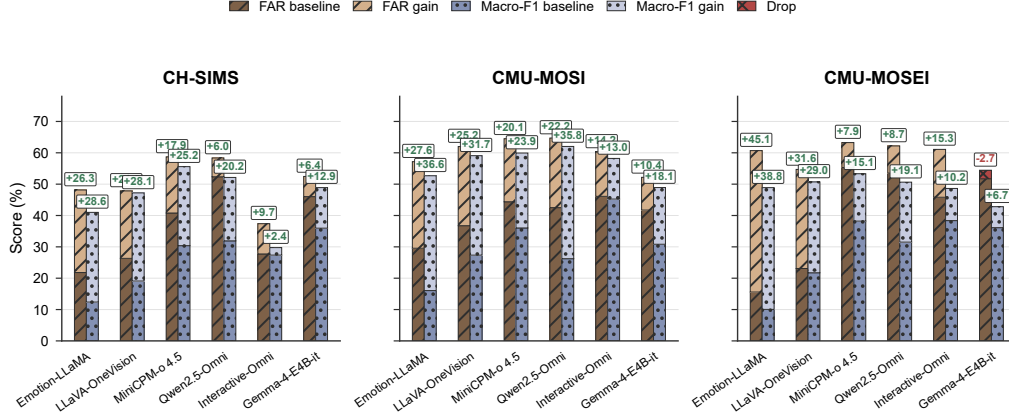


Figure 13: Supplementary visualization of FAR and Macro-F1 results in Table 1. Gains are shown relative to the corresponding zero-shot MLLM backbone.

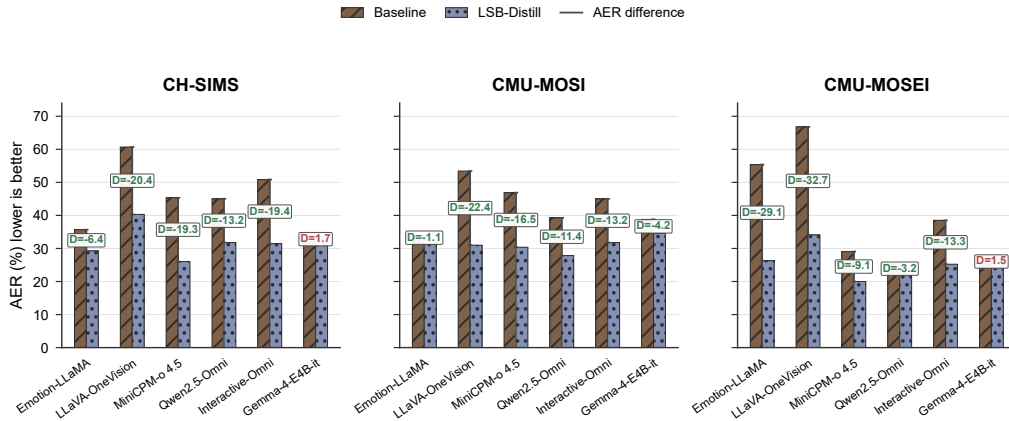


Figure 14: Supplementary visualization of AER results in Table 1. Lower AER indicates fewer adjacent-category mistakes.

766 **Existing assets and licenses.** We use CH-SIMS v2, CMU-MOSI, and CMU-MOSEI through their
767 official academic releases and cite the original dataset papers. CMU-MOSI and CMU-MOSEI are
768 listed as CC-BY-NC-4.0 research datasets, while the official CH-SIMS v2 repository provides the
769 dataset for academic use but does not list a separate SPDX license on the release page. For model
770 assets, Qwen2.5-Omni, MiniCPM-o 4.5, LLaVA-OneVision, and Gemma-4-E4B-it are released under
771 Apache-2.0; Interactive-Omni is released under MIT; and Emotion-LLaMA code is released under
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773 release will cite these assets, keep their notices, and will not redistribute restricted raw datasets or
774 third-party checkpoints.

775 C Additional Experiments

776 C.1 Supplementary Visualizations

777 Figure 13 visualizes the FAR and Macro-F1 results from Table 1. LSB-Distill improves most
778 backbone-dataset pairs, with especially clear gains on CMU-MOSI and CMU-MOSEI. Figure 14
779 shows the corresponding AER comparison. The overall pattern is consistent with the main results:
780 LSB-Distill usually moves the model toward lower adjacent-error rates, although a few already
781 low-AER settings show small increases.

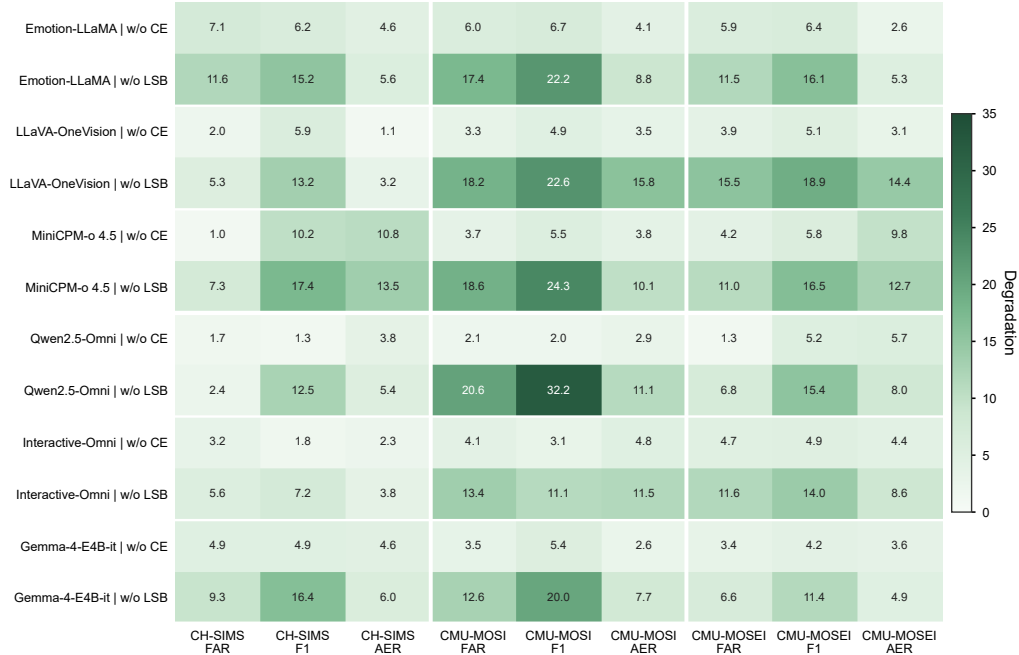


Figure 15: Degradation heatmap for the objective ablation in Table 2. Darker cells indicate larger performance loss or AER increase relative to the full LSB-Distill objective.

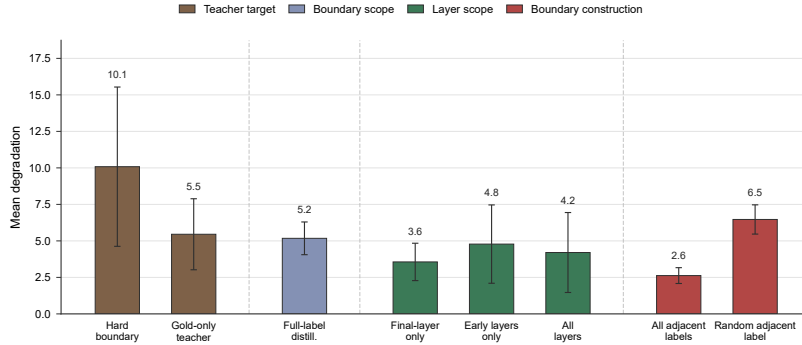


Figure 16: Average degradation of internal-module alternatives in Table 3. Error bars summarize variation across representative backbone-dataset pairs.

Figure 15 provides a compact view of the objective ablation in Table 2. Removing LSB causes larger degradation than removing CE in most settings, especially on CMU-MOSI, confirming that latent boundary supervision is central to the observed gains. Figure 16 summarizes the internal-module ablation from Table 3. The reference-conditioned teacher target and student-adaptive boundary construction are the most sensitive design choices, while full-label distillation and non-middle-to-late layer scopes are also weaker than the full setting.

C.2 Hyperparameter Sensitivity

We study hyperparameter sensitivity on Qwen2.5-Omni + CH-SIMS. We vary the LSB loss weight λ and the temperature τ , while keeping other training settings fixed. In the λ sweep, we set $\tau = 1.0$. In the τ sweep, we use the default LSB setting and change only the teacher-boundary temperature. Table 7 reports the results.

The λ sweep shows that stronger boundary supervision is not always better. A moderate weight works best: $\lambda = 1$ gives the highest FAR and F1. When λ becomes too large, the model puts more pressure

Table 7: Hyperparameter sensitivity of LSB-Distill on Qwen2.5-Omni / CH-SIMS. FAR and F1 are percentages; AER denotes Adjacent Error Rate. Best results within each block are in bold and second-best results are underlined.

Setting	FAR $\uparrow$	F1 $\uparrow$	AER $\downarrow$
<i>LSB weight λ</i>			
$\lambda = 1$	59.40	55.03	31.76
$\lambda = 2.5$	55.88	39.29	30.75
$\lambda = 5.0$	<u>58.35</u>	<u>52.14</u>	31.79
$\lambda = 7.5$	56.88	39.39	<u>30.45</u>
$\lambda = 10$	58.39	46.15	28.24
<i>Temperature τ</i>			
$\tau = 0.5$	56.72	48.60	32.85
$\tau = 0.75$	58.20	51.90	31.95
$\tau = 1.0$	58.35	52.14	31.79
$\tau = 1.5$	61.01	55.69	29.95
$\tau = 2.0$	<u>59.10</u>	<u>53.40</u>	<u>30.80</u>

on local boundary separation. This helps reduce adjacent errors, as seen with $\lambda = 10$, but it also hurts Macro-F1. In other words, too much boundary loss can weaken global label discrimination.

Temperature controls how sharp the teacher boundary target is. A small value such as $\tau = 0.5$ makes the target too sharp and gives weaker results. The best setting is $\tau = 1.5$, which improves FAR, F1, and AER over the default $\tau = 1.0$. This suggests that a mildly smoothed teacher target preserves useful boundary uncertainty. When τ increases to 2.0, performance drops slightly, likely because the target becomes too smooth and loses boundary contrast.

D Limitations

This work studies fine-grained multimodal sentiment analysis in an ordered five-label FAR space, which is useful for analyzing adjacent-category errors but still simplifies real affective expression. Mixed emotions, sarcasm, speaker context, and multi-dimensional emotion labels may require richer annotation schemes. Our experiments use CH-SIMS, CMU-MOSI, and CMU-MOSEI, so broader language, domain, and recording conditions remain to be tested; mapping CMU-MOSI and CMU-MOSEI into the unified five-label space also loses some intensity information. In addition, LSB-Distill needs access to candidate scores and intermediate hidden states during training, making it easier to apply to open-source MLLMs than to closed-source API models. The teacher mode adds training cost, although inference uses only the student mode. Finally, we focus mainly on adjacent-boundary errors, while non-adjacent mistakes and cross-domain robustness remain open questions.

E Broader Impact

This work is intended to improve the reliability of fine-grained multimodal sentiment analysis, which may benefit affect-aware dialogue systems, user feedback analysis, and multimodal interaction research. At the same time, sentiment inference can be sensitive: if used in hiring, education, healthcare, surveillance, or other high-stakes settings, errors or dataset bias could harm individuals or groups. We therefore view LSB-Distill as a research method for analyzing and improving model behavior, not as a standalone tool for high-stakes decision making. Any deployment should include privacy protection, domain-specific validation, and bias testing.

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